

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PHY 221 — Past Questions & Materials
by Renance

Course: PHY 221
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 5
Levels: 200

This study pack was compiled and restructured by Renance for PHY 221 students at Federal University of Petroleum Resources, Effurun. It brings together 5 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

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Paper 1 of 5 — PHY 221 - 2016/2017 - Second Semester - Thermal Physics

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



College : Science
 Department: Physics
 Semester: 2nd Semester
 Session : 2016/2017
 Course: Thermal Physics (2Unit)
 Course Code: PHY 221
 Instruction: Answer any three (3) questions in all (Time allowed: 2hrs)
 Date: 16-08-2017

Under-listed Constant is to be used appropriately.

Avogadro's constant = $6.022 \times 10^{23} \text{ mol}^{-1}$

QUESTION ONE

- (a) Explain two types of State parameters
- (b) State third law of thermodynamics
- (c) Ideal gas originally at T_1 & P_1 is compressed from $V_1 \rightarrow V_2 = \frac{V_1}{2}$. Temperature varied at each instance $P = AV$ where A constant.
- (i) Draw process on PV diagram
- (ii) Find T_2 in terms of T_1
- (iii) Find workdone by gas in terms of nR & T_1

QUESTION TWO

- (a) State 2nd law of thermodynamics in term of Entropy.
- (b) Find the change in entropy that results when a 2.30kg block of ice melts Slowly (reversibly) at 0°C (273K)
- (c) Explain the following terms (i) Isothermal process
- (ii) Adiabatic process
- (iii) Isobaric process

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QUESTION THREE

- (a) State the four postulates of thermodynamics
- (b) (i) Explain (ii) Adiabatic wall
(iii) Diathermal wall
- (c) Fresh air is primarily a mixture of Nitrogen N_2 (molecular mass ≈ 28.0 u), Oxygen O_2 (molecular mass ≈ 32.0 u) and argon Ar (atomic mass ≈ 39.9 u). Assume that each gas behaves as an Ideal gas.
At a temperature of 293K.
Find (a) the average translational kinetic energy
(b) the rms speed for each type of molecules.

QUESTION FOUR

- (a) State Zeroth law of Thermodynamics
- (b) An ideal gas undergoes an adiabatic reversible expansion from an initial state (T_1, V_1) to a final state (T_2, V_2) . Show that $\ln\left(\frac{T_2}{T_1}\right) = (\gamma - 1) \ln\left(\frac{V_1}{V_2}\right)$
- (c) An ideal Carnot heat pump is used to heat a house to a temperature of 21°C . How much work must be done by the pump to deliver 3350 J of heat into the house when the outdoor temperature is (a) 0°C
(b) -21°C

QUESTION FIVE

- (a) State Carnot's Principle as an alternative statement of the Second law of thermodynamics
- (b) Explain the two thermodynamics systems
- (c) Suppose the temperature of three moles of monatomic ideal gas is raised from $T_1 = 350\text{K}$ to $T_2 = 540\text{K}$ by two different methods. In the first method, 5500J of heat energy are added to the gas, while in the second method, 1500J are added in each case. Determine (a) the change in internal energy
(b) the work done

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Paper 2 of 5 — PHY 221 - 2015/2016 - Second Semester - Thermal Physics

Level 200 • Second Semester • PastQuestion

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DE) ARTIV NT OF PHYSICS

Seco d Sen uster Examination

Academic Session: (2015/2016

Course Code: PHY 221 (Therma Physic «)

Units:. 2 Units

Time Allowed: 2 hours

Instruction Answer question ne (1) and any other two (2) questions 09/09/2016.

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(i) NMP (ii) NBP. iii) NSP {iv) Triple Point of water

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{iiipExpiain any two of the heat transyer {mentioned in question 3ii)

Al) 'A solid wood door as dime} fons 2: 1 by 0.8m by fom 'and a thermal conductivity K-of 0.04 W/S/m/°C. The inside air tem) srature is 40°C and the outside air temperaturé is -10°C. "Assuming the gurface temperailure are hose of the Surrounding air, what is the rate of heat flow through the door, i

4. (a). List and exrialn'the three essantal pats of heat engine, a

46) fan ideal gas in'a cylinder ingially at a temperature of 17°C and.a pressure of $1.01 \times 10^5 \text{ N/m}^2$ is to be compressed to ong-cight of jts volume, What would be, the final pressure if the compression were cone (i) Isathermally {I} Adiabatically

(il) What would be the finat temperature in the Adiabatic diatomic system? (Ratio of molar heat.capacity is 2.40).

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Paper 3 of 5 — PHY 221 - 2016/2017 - Second Semester - Thermal Physics

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN



College : Science
 Department: Physics
 Semester: 2nd Semester
 Session : 2016/2017
 Course: Thermal Physics (2Unit)
 Course Code: PHY 221
 Instruction: Answer any three (3) questions in all (Time allowed: 2hrs)
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Seco d Sen uster Examination F d

Academic Session: 2015/2016

Course Code: PHY 221 (Therma Physic +)

Units:, 2 Units

Time Allowed: 2 hours

Instruction: Answer question ne (1) and any other two (2) questions 09/09/2016.

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7) -Asolid wood door has dime! ions 2) 1 by 0.8m by 4cm and a thermal conductivity K- of 0.04

WfS/m/°C. The inside air temperature is 20°C and the outside air temperature is -10°C, Assuming the surface temperature @ are those of the surrounding air, what is the rate of heat flow through the door, © i

4. (a) List and explain the three essential parts of heat engine. a

Ab) if an ideal gas in a cylinder initially at a temperature of 47°C and a pressure of $4.01 \times 10^5 \text{ N/m}^2$ is to be compressed to one-eighth of its volume, What would be, the final pressure if the compression were done (i) isothermally {ii) Adiabatically
(iit) What would be the final temperature in the Adiabatic diatomic system? (Ratio of molar heat, capacity is 1.40). a e F .

5. (a) For, isothermal expansion, q is absorbed and for isothermal compression q is rejected to the sink. Using Carnot Clausius Weoren show that
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Paper 5 of 5 — PHY 221 2010 2011 EXAM

Level 200 • Second Semester • PastQuestion

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FEDERAL UNIVERSITY OF PETROLEUM RESOURCES EFFURUN-NIGERIA
DEPARTMENT OF PHYSICS
B.Sc. (PHYSICS) DEGREE EXAMINATION 2010/2011 SESSION
PHY 221: THERMAL PHYSICS

Date: 27 January, 2012. Time Allowed: 2 Hours

INSTRUCTIONS: Answer ALL questions in section A and any TWO questions in section B. Answers to questions must be clear and concise. Credit will be given for useful diagrams to illustrate answers. Except if otherwise stated, the quantities and notations in equations retain their usual meanings.

SECTION A

a. State Nernst heat theorem

b. A sample of an ideal gas at 1 atm pressure expands reversibly and adiabatically from 1.5 dm^3 to 4.5 dm^3 with an initial temperature of 25°C . Calculate:

- its final temperature
- its final pressure
- the change in entropy

c. Explain the following pair of concepts

- relaxation time
- heat capacities
- cyclic process

d. An ideal monoatomic gas initially at 298 K and a pressure of 5 atmosphere is expanded to a final pressure of 1 atmosphere under the following conditions

- adiabatically and reversibly
- adiabatically and irreversibly

Calculate for each of the condition

- the final temperature of the gas
- the work done by the gas
- the heat absorbed by the gas
- the change in internal energy of the gas
- the change in the enthalpy of the gas

e. Calculate the standard entropy change at 25°C for the reaction

$$\text{H}_2(\text{g}) + \text{CO}_2(\text{g}) = \text{H}_2\text{O}(\text{g}) + \text{CO}(\text{g})$$

Given the following standard entropies

	H_2	CO_2	H_2O	CO
$S_{298}^\circ (\text{JK}^{-1}\text{mol}^{-1})$	130.5	213.8	188.7	197.9

f. State the Carnot theorem. A refrigerator in the laboratory rejects heat of about 27°C to its surrounding but operates with its cold reservoir at temperature -5°C . Assuming 200 g of H_2O melt to ice at 0°C , calculate the total work done by the motor of the refrigerator given that the maximum theoretical efficiency of ice (L_f) is $3.3 \times 10^3 \text{ JKg}^{-1}$

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SECTION B

- 1a The Gibb's free energy of a system is given by the equation
- $$G = H - TS$$
- Where H is the enthalpy, T the temperature and S the entropy. Use the other thermodynamic potentials to show that
- the work done per temperature increases at constant pressure
 - for an irreversible process at constant temperature and volume, the energy decreases until equilibrium is achieved.
- 1b. Distinguish clearly between state and path functions
- 1c. State four characteristics of entropy
- 2a. Derive an equation to show that any change in the entropy of a system is independent on the change in pressure of that system
- 2b. A monatomic gas has a volume of 80 m^3 at stp. Assuming the specific heat capacities at constant pressure and volume are $1.58 \text{ kJ kg}^{-1} \text{ K}^{-1}$ and $1.20 \text{ kJ kg}^{-1} \text{ K}^{-1}$ respectively, calculate
- the final temperature and pressure
 - the work done on the gas
- 2c. Write down the general equations for the internal energy and enthalpy under the following processes
- adiabatic
 - isochoric
 - ideal gas
- 3ai Briefly explain the term 'equilibrium state'
- 3aii State any three conditions for thermodynamic equilibrium
- 3b Show that the fundamental kinetic equation can expressed as
- $$pv = \frac{1}{3} Nmu^2$$
- Stating clearly all assumptions made
- 3c. Outline any three significance of the third law of thermodynamics
- 4a. Briefly discuss the concept of 'Carnot cycle' as used in the study of thermodynamics
- 4b. State the Zeroth law of thermodynamics bringing out its implications
- 4ci A gas has 10^{20} atom at a temperature of 57°C . Calculate the
- average kinetic energy of the molecule
 - total energy of the gas (take $K = 1.38 \times 10^{-23} \text{ J mol}^{-1} \text{ K}^{-1}$)
- 4cii Calculate the change in phase from liquid water to vapour at 100°C . the latent heat of vapourization is $22.6 \times 10^5 \text{ J/Kg}$. Assuming the vapour pressure at 100°C is 1 atm and V_g and V_f are $1.8 \text{ m}^3/\text{kg}$ and $10^{-3}/\text{kg}$, calculate the work done in pushing back the atmosphere to make room for the vapour.

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